

Отчёт по лабораторной работе 7

Адресация IPv4 и IPv6. Настройка DHCP

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Цель работы

Получение навыков настройки службы DHCP на сетевом оборудовании для распределения адресов IPv4 и IPv6.

Выполнение лабораторной работы

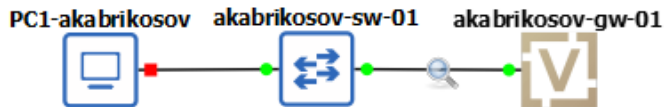
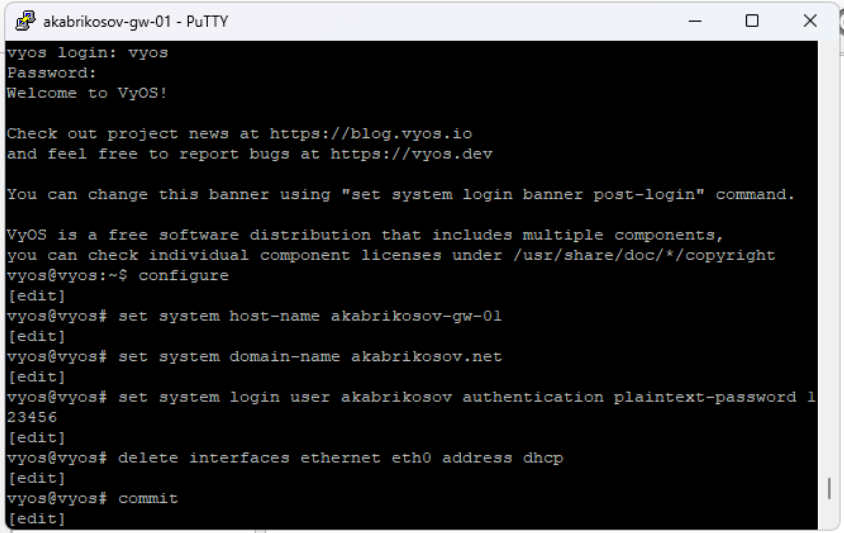


Рис. 1: Топология лабораторного стенда в GNS3



```
akabrikosov-gw-01 - PuTTY
vyos login: vyos
Password:
Welcome to VyOS!

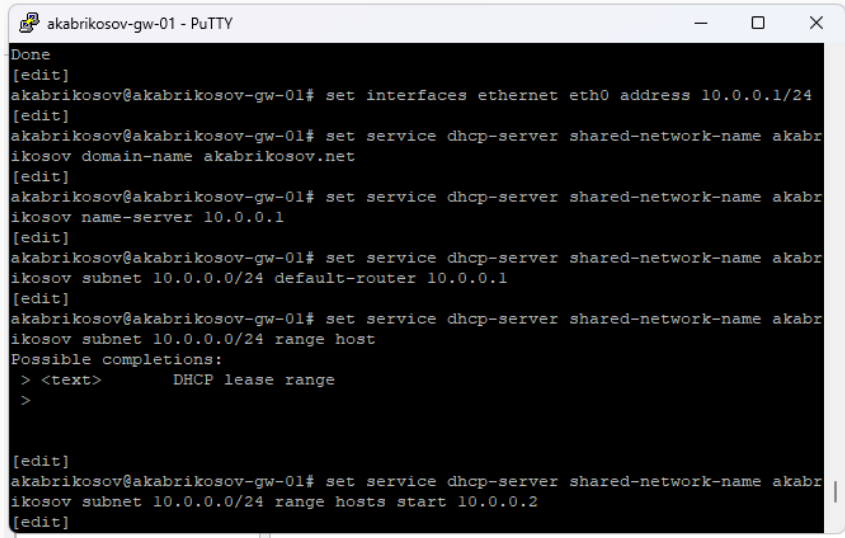
Check out project news at https://blog.vyos.io
and feel free to report bugs at https://vyos.dev

You can change this banner using "set system login banner post-login" command.

VyOS is a free software distribution that includes multiple components,
you can check individual component licenses under /usr/share/doc/*/copyright
vyos@vyos:~$ configure
[edit]
vyos@vyos# set system host-name akabrikosov-gw-01
[edit]
vyos@vyos# set system domain-name akabrikosov.net
[edit]
vyos@vyos# set system login user akabrikosov authentication plaintext-password 1
23456
[edit]
vyos@vyos# delete interfaces ethernet eth0 address dhcp
[edit]
vyos@vyos# commit
[edit]
```

Рис. 2: Настройка IPv4-адреса интерфейса eth0

Настройка DHCP-сервера (IPv4)



```
akabrikosov-gw-01 - PuTTY
Done
[edit]
akabrikosov@akabrikosov-gw-01# set interfaces ethernet eth0 address 10.0.0.1/24
[edit]
akabrikosov@akabrikosov-gw-01# set service dhcp-server shared-network-name akabrikosov domain-name akabrikosov.net
[edit]
akabrikosov@akabrikosov-gw-01# set service dhcp-server shared-network-name akabrikosov name-server 10.0.0.1
[edit]
akabrikosov@akabrikosov-gw-01# set service dhcp-server shared-network-name akabrikosov subnet 10.0.0.0/24 default-router 10.0.0.1
[edit]
akabrikosov@akabrikosov-gw-01# set service dhcp-server shared-network-name akabrikosov subnet 10.0.0.0/24 range host
Possible completions:
> <text>          DHCP lease range
>

[edit]
akabrikosov@akabrikosov-gw-01# set service dhcp-server shared-network-name akabrikosov subnet 10.0.0.0/24 range hosts start 10.0.0.2
[edit]
```

Получение адреса на клиенте (PC1)

```
PC1-akabrikosov - PuTTY
68:00
Option 12: Host Name = VPCS

Opcode: 2 (REPLY)
Client IP Address: 10.0.0.2
Your IP Address: 10.0.0.2
Server IP Address: 0.0.0.0
Gateway IP Address: 0.0.0.0
Client MAC Address: 00:50:79:66:68:00
Option 53: Message Type = Ack
Option 54: DHCP Server = 10.0.0.1
Option 51: Lease Time = 86400
Option 1: Subnet Mask = 255.255.255.0
Option 3: Router = 10.0.0.1
Option 6: DNS Server = 10.0.0.1
Option 15: Domain = akabrikosov.net

IP 10.0.0.2/24 GW 10.0.0.1

VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS>
```


Проверка связности (IPv4)

```
PC1-akabrikosov - PuTTY
Saving startup configuration to startup.vpc
. done

VPCS>
VPCS> show ip

NAME       : VPCS[1]
IP/MASK     : 10.0.0.2/24
GATEWAY     : 10.0.0.1
DNS         : 10.0.0.1
DHCP SERVER : 10.0.0.1
DHCP LEASE  : 86377, 86400/43200/75600
DOMAIN NAME : akabrikosov.net
MAC         : 00:50:79:66:68:00
LPORT      : 10010
RHOST:PORT  : 127.0.0.1:10011
MTU         : 1500

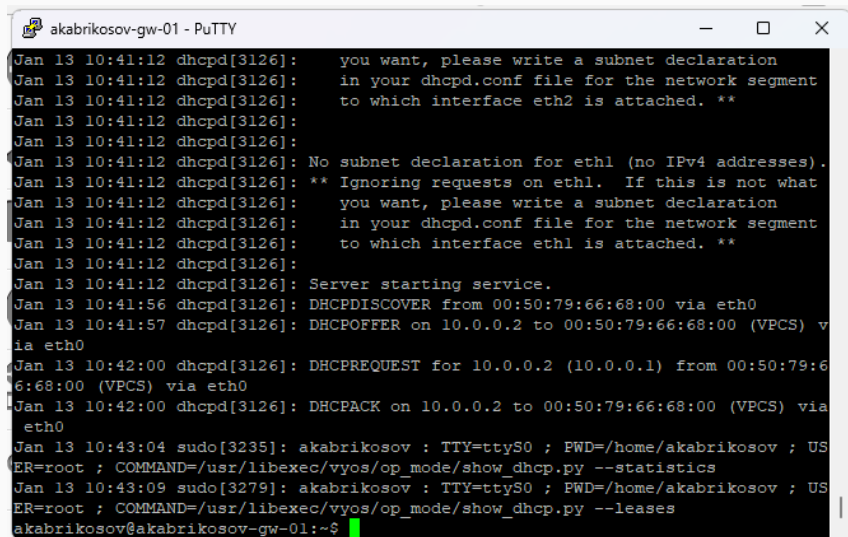
VPCS> ping 10.0.0.1 -c 2

84 bytes from 10.0.0.1 icmp_seq=1 ttl=64 time=1.084 ms
84 bytes from 10.0.0.1 icmp_seq=2 ttl=64 time=4.099 ms

VPCS> █
```

```
akabrikosov@akabrikosov-gw-01:~$  
akabrikosov@akabrikosov-gw-01:~$ show dhcp server statistics  
Pool           Size      Leases    Available  Usage  
-----  
akabrikosov     252        1         251       0%  
akabrikosov@akabrikosov-gw-01:~$ show dhcp server leases  
IP address      Hardware address  State      Lease start      Lease expiration  
  Remaining      Pool              Hostname  
-----  
10.0.0.2        00:50:79:66:68:00 active      2026/01/13 10:42:00 2026/01/14 10:42:00  
:00 23:58:50      akabrikosov VPCS  
akabrikosov@akabrikosov-gw-01:~$
```

Рис. 6: Статистика DHCP-сервера



```
akabrikosov-gw-01 - PuTTY
Jan 13 10:41:12 dhcpd[3126]: you want, please write a subnet declaration
Jan 13 10:41:12 dhcpd[3126]: in your dhcpd.conf file for the network segment
Jan 13 10:41:12 dhcpd[3126]: to which interface eth2 is attached. **
Jan 13 10:41:12 dhcpd[3126]:
Jan 13 10:41:12 dhcpd[3126]: No subnet declaration for eth1 (no IPv4 addresses).
Jan 13 10:41:12 dhcpd[3126]: ** Ignoring requests on eth1. If this is not what
Jan 13 10:41:12 dhcpd[3126]: you want, please write a subnet declaration
Jan 13 10:41:12 dhcpd[3126]: in your dhcpd.conf file for the network segment
Jan 13 10:41:12 dhcpd[3126]: to which interface eth1 is attached. **
Jan 13 10:41:12 dhcpd[3126]:
Jan 13 10:41:12 dhcpd[3126]: Server starting service.
Jan 13 10:41:56 dhcpd[3126]: DHCPDISCOVER from 00:50:79:66:68:00 via eth0
Jan 13 10:41:57 dhcpd[3126]: DHCPOFFER on 10.0.0.2 to 00:50:79:66:68:00 (VPCS) v
ia eth0
Jan 13 10:42:00 dhcpd[3126]: DHCPREQUEST for 10.0.0.2 (10.0.0.1) from 00:50:79:6
6:68:00 (VPCS) via eth0
Jan 13 10:42:00 dhcpd[3126]: DHCPACK on 10.0.0.2 to 00:50:79:66:68:00 (VPCS) via
eth0
Jan 13 10:43:04 sudo[3235]: akabrikosov : TTY=ttyS0 ; PWD=/home/akabrikosov ; US
ER=root ; COMMAND=/usr/libexec/vyos/op_mode/show_dhcp.py --statistics
Jan 13 10:43:09 sudo[3279]: akabrikosov : TTY=ttyS0 ; PWD=/home/akabrikosov ; US
ER=root ; COMMAND=/usr/libexec/vyos/op_mode/show_dhcp.py --leases
akabrikosov@akabrikosov-gw-01:~$
```

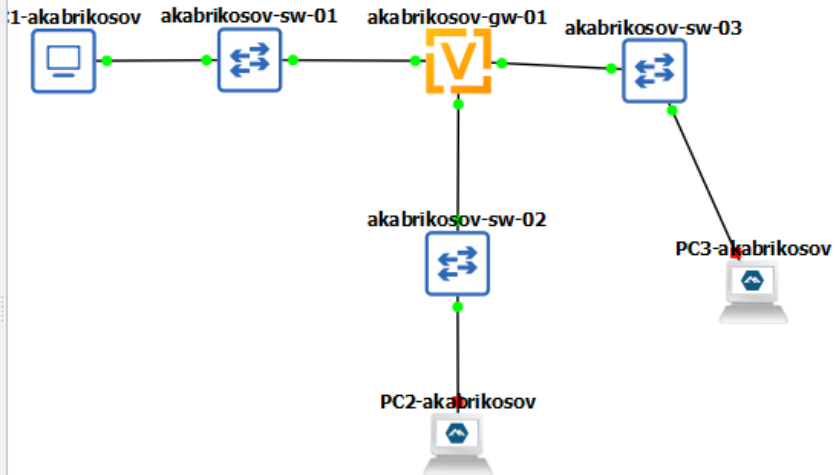
Рис 7: Журнал работы DHCP-сервера

Анализ захваченного трафика (DHCP)

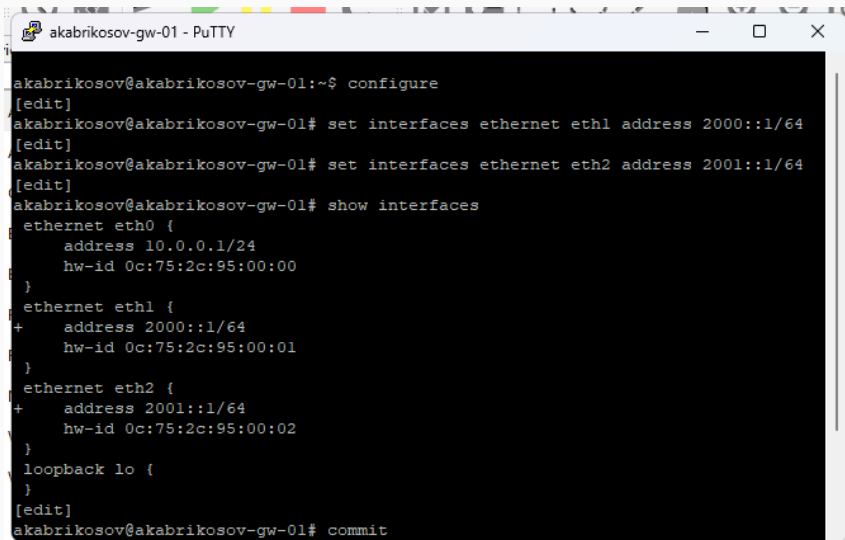
No.	Time	Source	Destination	Protocol	Length	Info
20	367.924360	0.0.0.0	255.255.255.255	DHCP	406	DHCP Discover - Transaction ID 0x78d8eb1
21	367.943756	0c:75:2c:95:00:00	Broadcast	ARP	60	Who has 10.0.0.2? Tell 10.0.0.1
22	368.924998	0.0.0.0	255.255.255.255	DHCP	406	DHCP Discover - Transaction ID 0x78d8eb1
23	368.946610	10.0.0.1	10.0.0.2	DHCP	342	DHCP Offer - Transaction ID 0x78d8eb1
24	369.000637	0c:75:2c:95:00:00	Broadcast	ARP	60	Who has 10.0.0.2? Tell 10.0.0.1
25	370.025144	0c:75:2c:95:00:00	Broadcast	ARP	60	Who has 10.0.0.2? Tell 10.0.0.1
26	371.925724	0.0.0.0	255.255.255.255	DHCP	406	DHCP Request - Transaction ID 0x78d8eb1
27	371.933418	10.0.0.1	10.0.0.2	DHCP	342	DHCP ACK - Transaction ID 0x78d8eb1

Client IP address: 10.0.0.2	0000
Your (client) IP address: 10.0.0.2	0010
Next server IP address: 0.0.0.0	0020
Relay agent IP address: 0.0.0.0	0030
Client MAC address: Private_66:68:00 (00:50:79:66:68:00)	0040
Client hardware address padding: 00000000000000000000	0050
Server host name not given	0060
Boot file name not given	0070
Magic cookie: DHCP	0080
✓ Option: (53) DHCP Message Type (ACK)	0090
Length: 1	00a0
DHCP: ACK (5)	00b0
✓ Option: (54) DHCP Server Identifier (10.0.0.1)	00c0
Length: 4	00d0
DHCP Server Identifier: 10.0.0.1	00e0
✓ Option: (51) IP Address Lease Time	00f0
Length: 4	0100
IP Address Lease Time: 1 day (86400)	0110
✓ Option: (1) Subnet Mask (255.255.255.0)	0120
Length: 4	0130
Subnet Mask: 255.255.255.0	0140
✓ Option: (3) Router	0150
Length: 4	

Расширение топологии (IPv6)



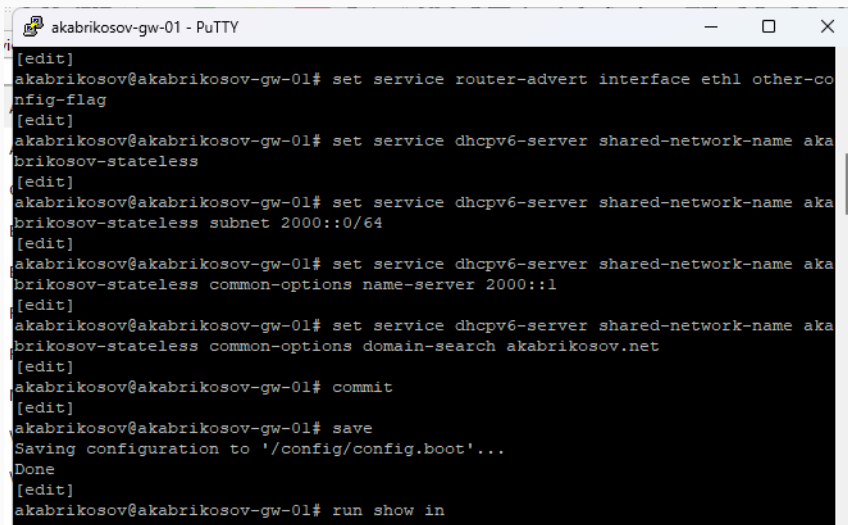
Настройка IPv6-адресов на VyOS



```
akabrikosov-gw-01 - PuTTY
akabrikosov@akabrikosov-gw-01:~$ configure
[edit]
akabrikosov@akabrikosov-gw-01# set interfaces ethernet eth1 address 2000::1/64
[edit]
akabrikosov@akabrikosov-gw-01# set interfaces ethernet eth2 address 2001::1/64
[edit]
akabrikosov@akabrikosov-gw-01# show interfaces
    ethernet eth0 {
        address 10.0.0.1/24
        hw-id 0c:75:2c:95:00:00
    }
    ethernet eth1 {
+   address 2000::1/64
        hw-id 0c:75:2c:95:00:01
    }
    ethernet eth2 {
+   address 2001::1/64
        hw-id 0c:75:2c:95:00:02
    }
    loopback lo {
    }
[edit]
akabrikosov@akabrikosov-gw-01# commit
```

Рис. 10: Настройка IPv6-адресов на интерфейсах VyOS

Настройка RA + DHCPv6 Stateless (eth1)



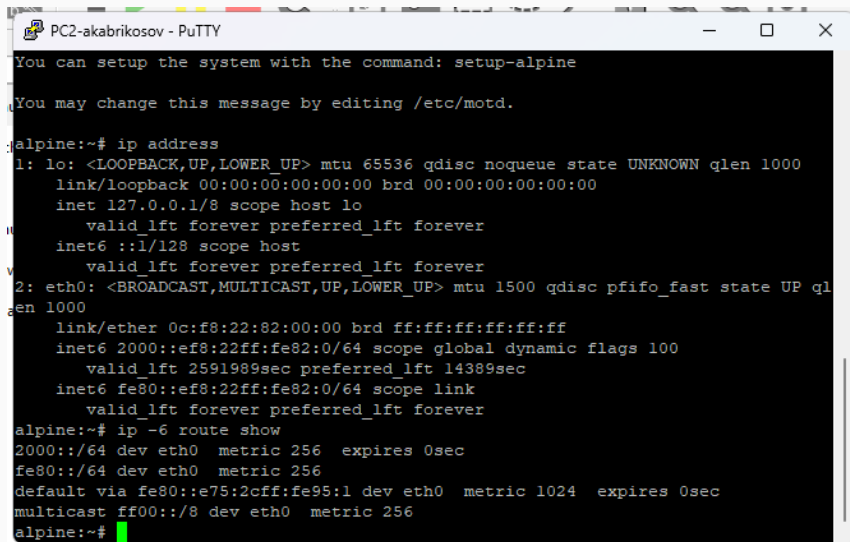
```
akabrikosov-gw-01 - PuTTY
[edit]
akabrikosov@akabrikosov-gw-01# set service router-advert interface eth1 other-co
nfig-flag
[edit]
akabrikosov@akabrikosov-gw-01# set service dhcpv6-server shared-network-name aka
brikosov-stateless
[edit]
akabrikosov@akabrikosov-gw-01# set service dhcpv6-server shared-network-name aka
brikosov-stateless subnet 2000::0/64
[edit]
akabrikosov@akabrikosov-gw-01# set service dhcpv6-server shared-network-name aka
brikosov-stateless common-options name-server 2000::1
[edit]
akabrikosov@akabrikosov-gw-01# set service dhcpv6-server shared-network-name aka
brikosov-stateless common-options domain-search akabrikosov.net
[edit]
akabrikosov@akabrikosov-gw-01# commit
[edit]
akabrikosov@akabrikosov-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
akabrikosov@akabrikosov-gw-01# run show in
```

Рис. 11: Настройка Router Advertisements и DHCPv6 Stateless

Проверка конфигурации сервисов на VyOS

```
akabrikosov-gw-01 - PuTTY
service {
  dhcp-server {
    shared-network-name akabrikosov {
      domain-name akabrikosov.net
      name-server 10.0.0.1
      subnet 10.0.0.0/24 {
        default-router 10.0.0.1
        range hosts {
          start 10.0.0.2
          stop 10.0.0.253
        }
      }
    }
  }
  dhcpv6-server {
    shared-network-name akabrikosov-stateless {
      common-options {
        domain-search akabrikosov.net
        name-server 2000::1
      }
      subnet 2000::0/64 {
      }
    }
  }
}
```


Проверка клиента PC2 (Stateless)



```
PC2-akabrikov - PuTTY
You can setup the system with the command: setup-alpine
You may change this message by editing /etc/motd.

alpine:~# ip address
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP qlen 1000
    link/ether 0c:f8:22:82:00:00 brd ff:ff:ff:ff:ff:ff
    inet6 2000::ef8:22ff:fe82:0/64 scope global dynamic flags 100
        valid_lft 2591989sec preferred_lft 14389sec
    inet6 fe80::ef8:22ff:fe82:0/64 scope link
        valid_lft forever preferred_lft forever
alpine:~# ip -6 route show
2000::/64 dev eth0 metric 256 expires 0sec
fe80::/64 dev eth0 metric 256
default via fe80::e75:2cff:fe95:1 dev eth0 metric 1024 expires 0sec
multicast ff00::/8 dev eth0 metric 256
alpine:~#
```

Проверка связности (IPv6)

```
alpine:~#  
alpine:~# ping 2000::1 -c 2  
PING 2000::1 (2000::1): 56 data bytes  
64 bytes from 2000::1: seq=0 ttl=64 time=2.894 ms  
64 bytes from 2000::1: seq=1 ttl=64 time=1.525 ms  
  
--- 2000::1 ping statistics ---  
2 packets transmitted, 2 packets received, 0% packet loss  
round-trip min/avg/max = 1.525/2.209/2.894 ms  
alpine:~# cat /etc/resolv.conf  
nameserver 10.0.2.3  
alpine:~#
```

Рис. 14: Ping IPv6-адреса маршрутизатора

Получение параметров по DHCPv6 (Stateless)

```
udhcp6: sending discover
udhcp6: sending select
udhcp6: no IAADDR option, ignoring packet
udhcp6: sending select
udhcp6: sending select
^C
alpine:~# ping 2000::1 -c 2
PING 2000::1 (2000::1): 56 data bytes
64 bytes from 2000::1: seq=0 ttl=64 time=1.375 ms
64 bytes from 2000::1: seq=1 ttl=64 time=2.610 ms

--- 2000::1 ping statistics ---
2 packets transmitted, 2 packets received, 0% packet loss
round-trip min/avg/max = 1.375/1.992/2.610 ms
alpine:~# cat /etc/resolv.conf
nameserver 10.0.2.3
alpine:~#
```

Рис. 15: Получение параметров по DHCPv6 Stateless

```
[edit]
akabrikosov@akabrikosov-gw-01# run show dhcpv6 server leases
IPv6 address      State      Last communication      Lease expiration      Remaining
Type      Pool      IAID_DUID
-----
n [edit]
S akabrikosov@akabrikosov-gw-01#
```

Рис. 16: Состояние аренд DHCPv6

Анализ трафика (Stateless DHCPv6)

No.	Time	Source	Destination	Protocol	Length	Info
47	126.576164	fe80::ef8:22ff:fe82...	fe80::e75:2cff:fe95...	ICMPv6	210	Destination Unreachable (Port unreachable)
48	126.692372	fe80::ef8:22ff:fe82...	ff02::1:2	DHCPv6	134	Request XID: 0x747813 [ROOT-ONLY DOMAIN NAM
49	126.694306	fe80::e75:2cff:fe95...	fe80::ef8:22ff:fe82...	DHCPv6	162	Reply XID: 0x747813 CID: 000300010cf82282000
50	126.695374	fe80::ef8:22ff:fe82...	fe80::e75:2cff:fe95...	ICMPv6	210	Destination Unreachable (Port unreachable)
51	129.782768	fe80::ef8:22ff:fe82...	ff02::1:2	DHCPv6	182	Request XID: 0x747813 [ROOT-ONLY DOMAIN NAM
52	129.789340	fe80::e75:2cff:fe95...	fe80::ef8:22ff:fe82...	DHCPv6	162	Reply XID: 0x747813 CID: 000300010cf82282000
53	129.789908	fe80::ef8:22ff:fe82...	fe80::e75:2cff:fe95...	ICMPv6	210	Destination Unreachable (Port unreachable)
54	131.602606	fe80::ef8:22ff:fe82...	fe80::e75:2cff:fe95...	ICMPv6	86	Neighbor Solicitation for fe80::e75:2cff:fe9

Length: 108
Checksum: 0x4351 [unverified]
[Checksum Status: Unverified]
[Stream index: 1]
[Stream Packet Number: 9]
> [Timestamps]
UDP payload (100 bytes)

▼ DHCPv6

Message type: Reply (7)
Transaction ID: 0x747813

> Identity Association for Non-temporary Address

▼ Client Identifier

Option: Client Identifier (1)
Length: 10
DUID: 000300010cf822820000
DUID Type: link-layer address (3)
Hardware type: Ethernet (1)
Link-layer address: 0c:f8:22:82:00:00
Link-layer address (Ethernet): 0c:f8:22:82:00:00 (0c:f8:22:82:00:00)

▼ Server Identifier

Option: Server Identifier (2)
Length: 14
DUID: 0001000130f8e07e0c752c950001

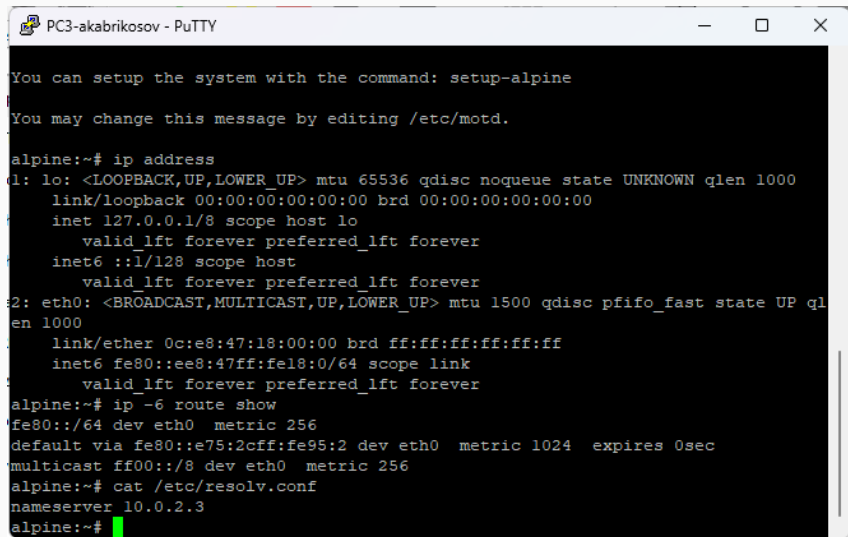
0000 0c f8
0010 2a 05
0020 2c f1
0030 22 f1
0040 78 13
0050 00 0e
0060 73 73
0070 6f 72
0080 65 2e
0090 00 02
00a0 00 01

Standard input: <live capture in progress> | Пакеты: 68 | Профиль: Default

Настройка RA + DHCPv6 Stateful (eth2)

```
akabrikosov-gw-01 - PuTTY
[edit]
akabrikosov@akabrikosov-gw-01# set service router-advert interface eth2 managed-
flag
[edit]
akabrikosov@akabrikosov-gw-01# set service dhcpv6-server shared-network-name aka
brikosov-stateful
[edit]
akabrikosov@akabrikosov-gw-01# set service dhcpv6-server shared-network-name aka
brikosov-stateful subnet 2001::0/64
[edit]
akabrikosov@akabrikosov-gw-01# set service dhcpv6-server shared-network-name aka
brikosov-stateful subnet 2001::0/64 name-server 2001::1
[edit]
akabrikosov@akabrikosov-gw-01# set service dhcpv6-server shared-network-name aka
brikosov-stateful subnet 2001::0/64 domain-search akabrikosov.net
[edit]
akabrikosov@akabrikosov-gw-01# set service dhcpv6-server shared-network-name aka
brikosov-stateful subnet 2001::0/64 address-range start 2001::100 stop 2001::199
[edit]
akabrikosov@akabrikosov-gw-01# commit
[edit]
akabrikosov@akabrikosov-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
akabrikosov@akabrikosov-gw-01#
```

Начальные настройки клиента PC3 (до DHCPv6)



```
PC3-akabrikosov - PuTTY

You can setup the system with the command: setup-alpine

You may change this message by editing /etc/motd.

alpine:~# ip address
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP ql
en 1000
    link/ether 0c:e8:47:18:00:00 brd ff:ff:ff:ff:ff:ff
    inet6 fe80::ee8:47ff:fe18:0/64 scope link
        valid_lft forever preferred_lft forever
alpine:~# ip -6 route show
fe80::/64 dev eth0  metric 256
default via fe80::e75:2cff:fe95:2 dev eth0  metric 1024  expires 0sec
multicast ff00::/8 dev eth0  metric 256
alpine:~# cat /etc/resolv.conf
nameserver 10.0.2.3
alpine:~#
```

Рис. 19: Начальные IPv6-настройки на PC3

Получение IPv6 по DHCPv6 Stateful (PC3)

```
PC3-akabrikosov - PuTTY
nameserver 10.0.2.3
alpine:~#
alpine:~# udhcp6 -i eth0
udhcp6: started, v1.36.1
udhcp6: sending discover
udhcp6: sending select
udhcp6: IPv6 obtained, lease time 43200
alpine:~# ip address
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP qlen 1000
    link/ether 0c:e8:47:18:00:00 brd ff:ff:ff:ff:ff:ff
    inet6 fe80::e8:47ff:fe18:0/64 scope link
        valid_lft forever preferred_lft forever
alpine:~# ip -6 route show
fe80::/64 dev eth0 metric 256
default via fe80::e75:2cff:fe95:2 dev eth0 metric 1024 expires 0sec
multicast ff00::/8 dev eth0 metric 256
alpine:~# ping 2001::1 -c 2
PING 2001::1 (2001::1): 56 data bytes
64 bytes from 2001::1: seq=0 ttl=64 time=2.877 ms
64 bytes from 2001::1: seq=1 ttl=64 time=2.069 ms

--- 2001::1 ping statistics ---
2 packets transmitted, 2 packets received, 0% packet loss
round-trip min/avg/max = 2.069/2.473/2.877 ms
alpine:~# cat /etc/resolv.conf
search akabrikosov.net
nameserver 2001:0000:0000:0000:0000:0000:0000:0001
alpine:~#
```



```
akabrikosov@akabrikosov-gw-01# run show dhcpv6 server leases
IPv6 address      State      Last communication    Lease expiration      Remaining
Type      Pool      IAID_DUID
-----
[edit]
akabrikosov@akabrikosov-gw-01# run show dhcpv6 server leases
IPv6 address      State      Last communication    Lease expiration      Remaining
Type      Pool      IAID_DUID
-----
2001::199         active    2026/01/13 11:02:11    2026/01/13 23:02:11    11:58:52
non-temporary    akabrikosov-stateful  20:8f:0d:e5:00:03:00:01:0c:e8:47:18:00:00
[edit]
akabrikosov@akabrikosov-gw-01#
```

Рис. 21: Список аренд DHCPv6

Анализ трафика (Stateful DHCPv6)

No.	Time	Source	Destination	Protocol	Length	Info
4	16.704800	fe80::e75:2cff:fe95...	ff02::1	ICMPv6	86	Router Advertisement from 0c:75:2c:95:00:02
5	32.739200	fe80::e75:2cff:fe95...	ff02::1	ICMPv6	86	Router Advertisement from 0c:75:2c:95:00:02
6	93.612949	fe80::ee8:47ff:fe18...	ff02::1:2	DHCPv6	116	Solicit XID: 0x8ea05a [ROOT-ONLY DOMAIN NAM
7	93.630650	fe80::e75:2cff:fe95...	fe80::ee8:47ff:fe18...	DHCPv6	183	Advertise XID: 0x8ea05a IAA: 2001::199 CID:
8	93.631737	fe80::ee8:47ff:fe18...	fe80::e75:2cff:fe95...	ICMPv6	231	Destination Unreachable (Port unreachable)
9	93.731110	fe80::ee8:47ff:fe18...	ff02::1:2	DHCPv6	134	Request XID: 0x8ea05a [ROOT-ONLY DOMAIN NAM
10	93.733698	fe80::e75:2cff:fe95...	fe80::ee8:47ff:fe18...	DHCPv6	183	Reply XID: 0x8ea05a IAA: 2001::199 CID: 0003
11	93.734117	fe80::ee8:47ff:fe18...	fe80::e75:2cff:fe95...	ICMPv6	231	Destination Unreachable (Port unreachable)

> Frame 10: Packet, 183 bytes on wire (1464 bits), 183 bytes captured (1464 bits) on interface -, id 0 > Ethernet II, Src: 0c:75:2c:95:00:02 (0c:75:2c:95:00:02), Dst: 0c:e8:47:18:00:00 (0c:e8:47:18:00:00) > Internet Protocol Version 6, Src: fe80::e75:2cff:fe95:2, Dst: fe80::ee8:47ff:fe18:0 > User Datagram Protocol, Src Port: 547, Dst Port: 546 ▼ DHCPv6 Message type: Reply (7) Transaction ID: 0x8ea05a > Identity Association for Non-temporary Address ▼ Client Identifier Option: Client Identifier (1) Length: 10 DUID: 000300010ce847180000 DUID Type: link-layer address (3) Hardware type: Ethernet (1) Link-layer address: 0c:e8:47:18:00:00 Link-layer address (Ethernet): 0c:e8:47:18:00:00 (0c:e8:47:18:00:00) > Server Identifier ▼ DNS recursive name server Option: DNS recursive name server (23) Length: 16 1 DNS server address: 2001::1 ▼ Domain Search List	0000 0c e8 0010 ac 5d 0020 2c ff 0030 47 ff 0040 a0 5a 0050 00 00 0060 00 00 0070 00 0a 0080 00 01 0090 00 10 00a0 00 01 00b0 6f 76
--	---

Итоги работы

В ходе лабораторной работы: - Развернут стенд в GNS3 с VyOS, коммутаторами и клиентами (VPCS и Kali Linux CLI). - Настроен DHCP для IPv4: - автоматическая выдача адресов, - передача маски/шлюза/DNS/домена, - подтверждена корректность по статистике, арендам, журналу и анализу трафика. - Реализованы два режима DHCPv6: - Stateless: адрес через SLAAC (RA), параметры через DHCPv6, без аренд адресов. - Stateful: адреса выдаются DHCPv6-сервером из диапазона, сервер ведёт учёт аренд. - Анализ захваченного трафика подтвердил корректные сценарии DHCP, DHCPv6, RA и ICMPv6, соответствующие требованиям лабораторной работы.